
CONCEPTUAL DATA MODELING: ENTITY RELATIONSHIP DIAGRAM (ERD)

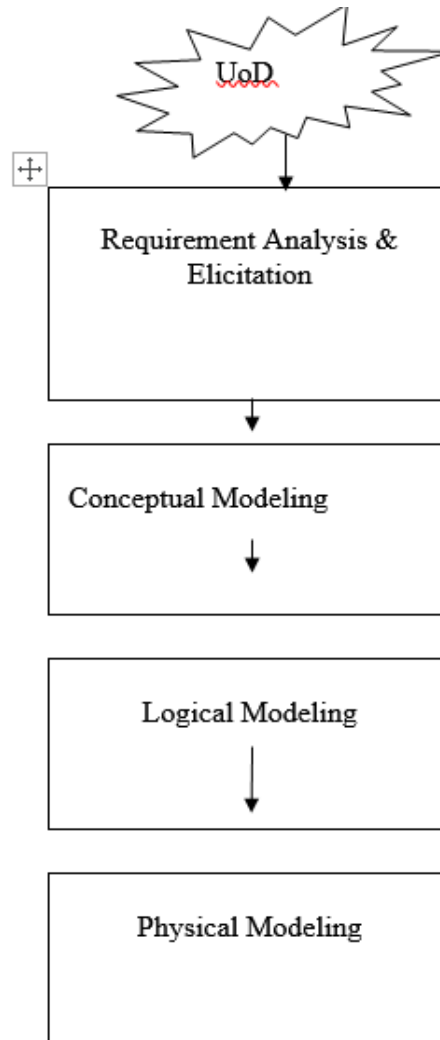
In This Lecture

- Entity/Relationship models
 - Entities and Attributes
 - Relationships
 - Attributes
 - E/R Diagrams

Approaches for data model development

- Top-down Approach
 - 3 steps
 - identify data entities
 - determine attributes of the entities
 - determine the nature of the relationships
 - usually results in a data model that is well organized but details can be easily overlooked.
- Bottom-up Approach
 - gather information on data used by the organization by...
 - group into entities of which these data are attributes
 - determine the nature of the relationships
 - insures that no important data is overlooked but overall organization may not be so apparent.

LEVELS IN DATA MODEL



Entity-Relationship Diagrams

- What Is an ERD?
 - A picture showing the information created, stored, and used by a business system.
 - ERD is network model that describes stored data of a system at a high level of abstraction.
 - Entities generally represent similar kinds of information
 - Lines drawn between entities show relationships among the data
 - High level business rules are also shown

Entity

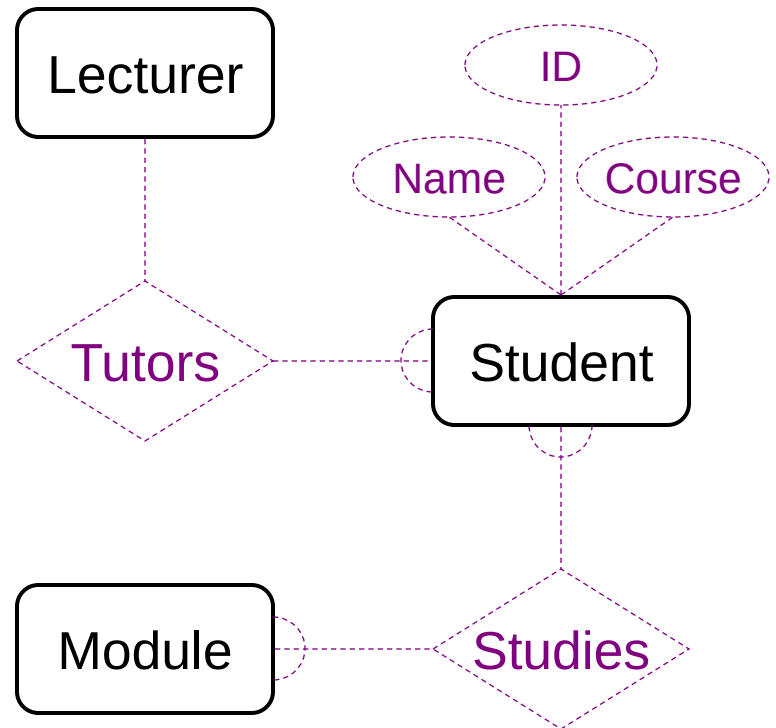
- Entities represent objects or things of interest
 - Physical things like students, lecturers, employees, products
 - More abstract things like modules, orders, courses, projects
 - noun
- Entities represent objects or things of interest
 - Physical things like students, lecturers, employees, products
 - More abstract things like modules, orders, courses, projects
 - noun

CUSTOMER

PART-TIME
EMPLOYEE

Diagramming Entities

- In an E/R Diagram, an entity is usually drawn as a box with rounded corners
- The box is labelled with the name of the class of objects represented by that entity

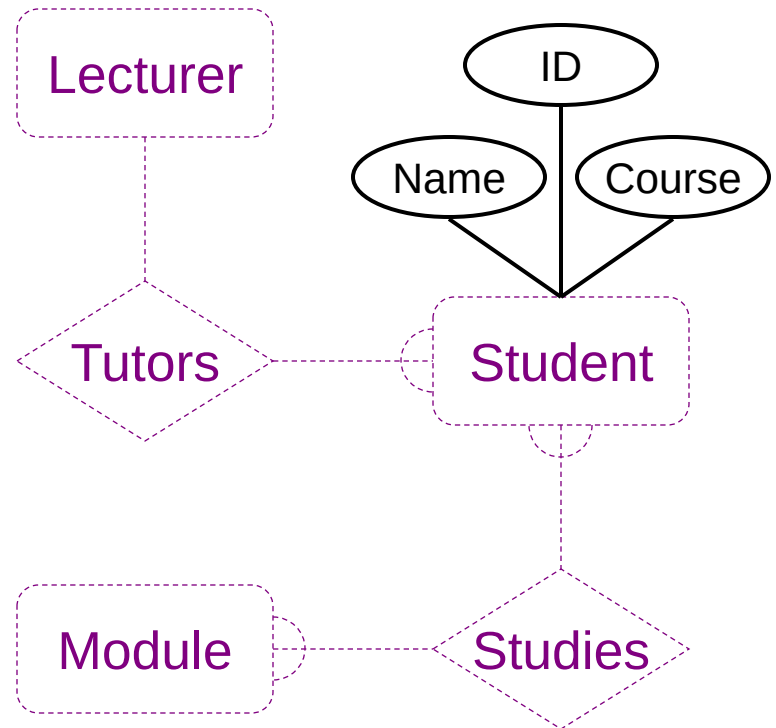


Attributes

- Attributes are facts, aspects, properties, or details about an entity
 - Students have IDs, names, courses, addresses, ...
 - Modules have codes, titles, credit weights, levels, ...
- Attributes have
 - A name
 - An associated entity
 - Domains of possible values
 - Values from the domain for each instance of the entity they are belong to

Diagramming Attributes

- In an E/R Diagram attributes may be drawn as ovals
- Each attribute is linked to its entity by a line
- The name of the attribute is written in the oval

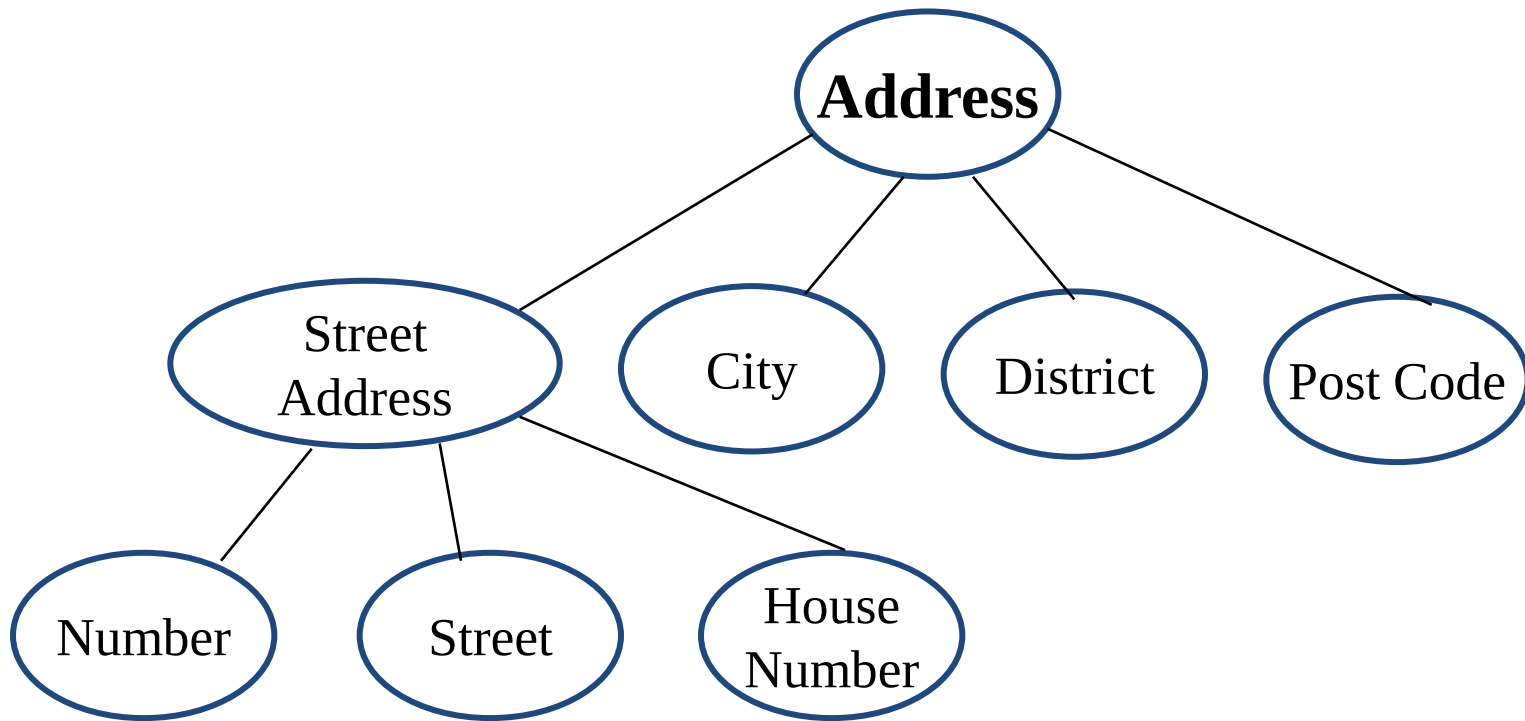


Types of Attributes

- Simple:
 - Each entity has a single atomic value for the attribute, for example SSN, CourseNo.
- Composite:
 - The attribute may be composed of several components
- Multivalued:
 - An entity may have multiple values for that attribute; for example Color of a Car or PreviousDegrees of a Student
- Derived:
 - The domain value of attribute can be determined from one or more other attributes.

Nested attribute

In general, composite and multiple-valued attributes may be nested arbitrarily to any number of levels although this is rare.

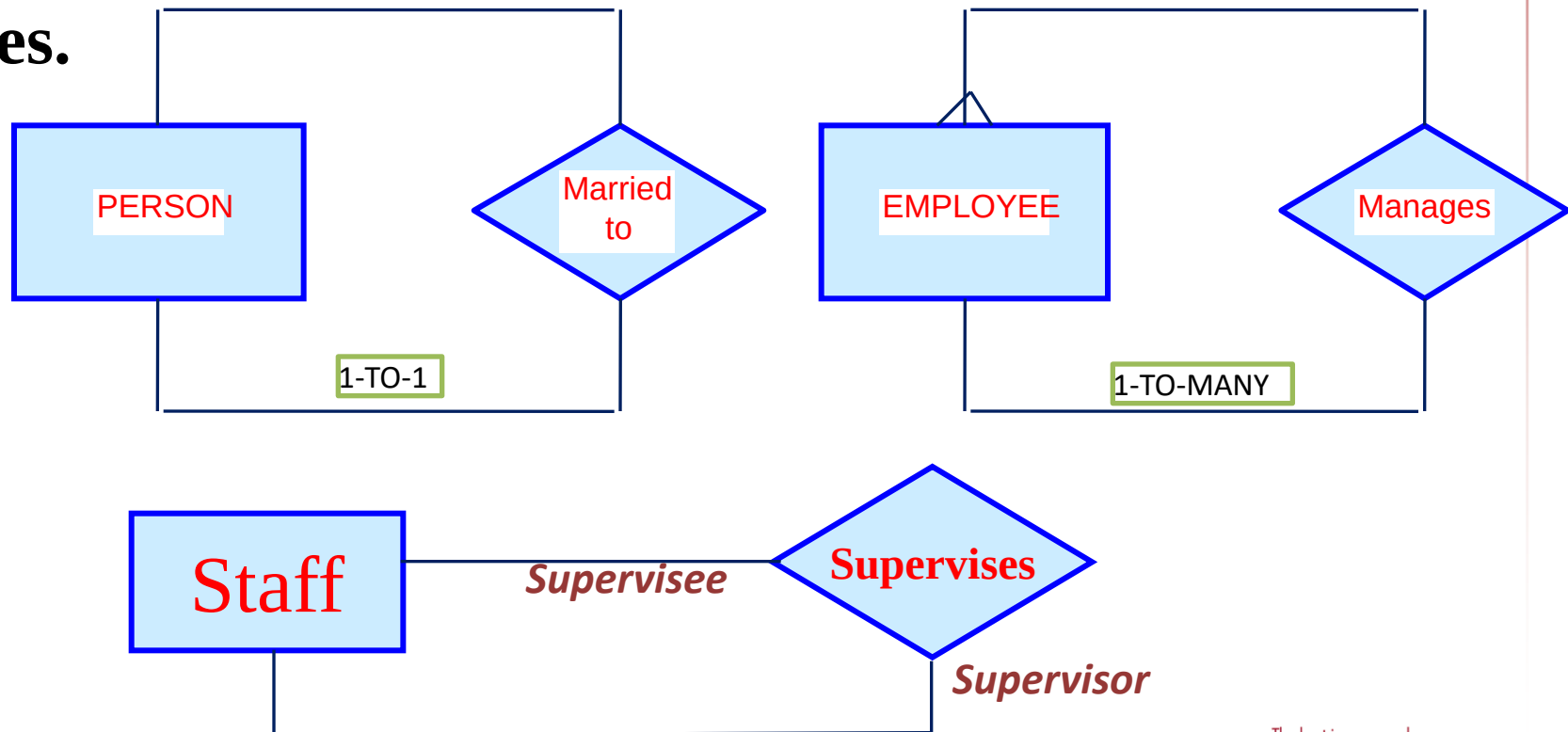


Relationships

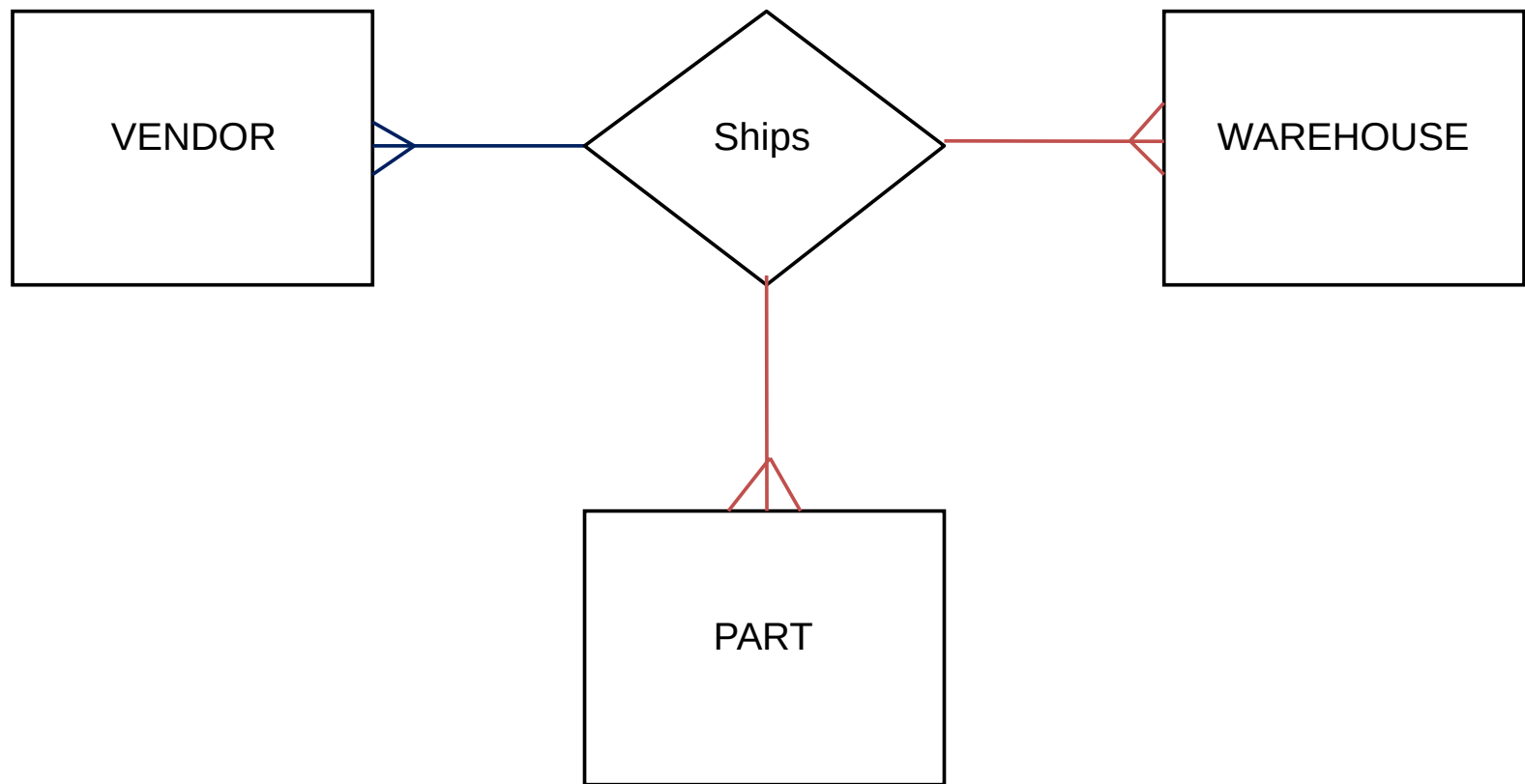
- Relationships are an association between two or more entities
 - Each Student takes several Modules
 - Each Module is taught by a Lecturer
 - Each Employee works for a single Department
- Relationships have
 - A name
 - A set of entities that participate in them
 - A degree - the number of entities that participate (most have degree 2)
 - A cardinality ratio

Unary or Recursive Relationships

A relationship where the same entity participates more than once in a different roles.



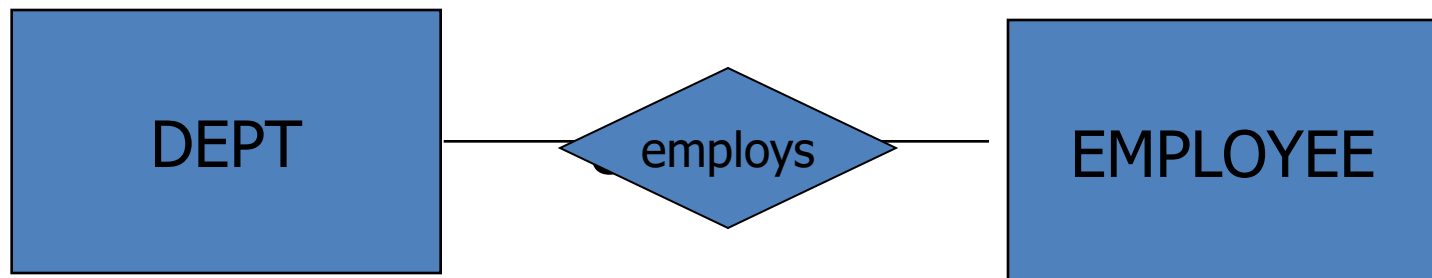
Ternary Relationships



Relationship

- A Relationship Type is a relevant business association between two Entity Types.
- A Relationship is an occurrence of a relationship type.

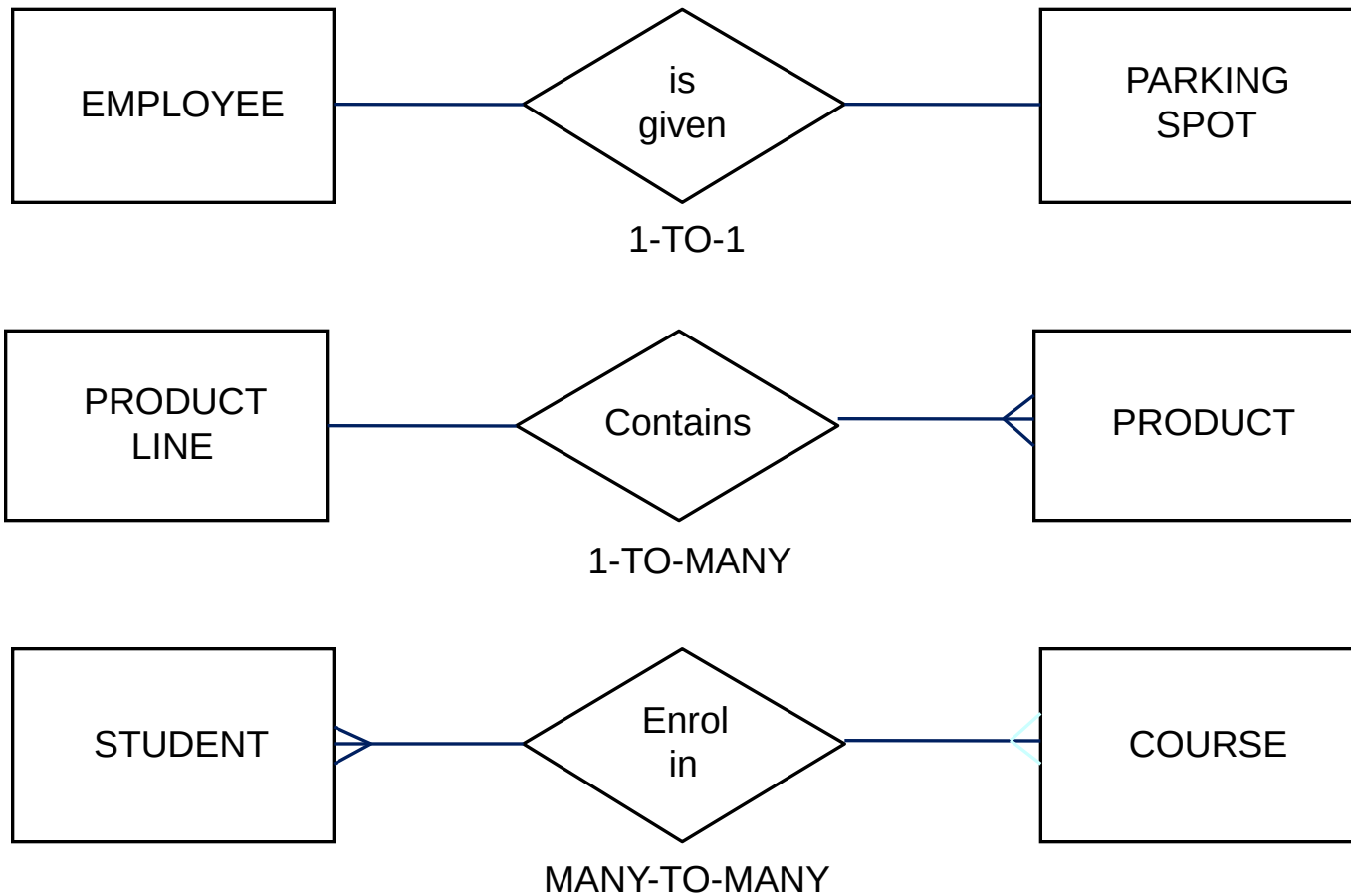
Marketing	_____	employs	_____	Yosef Ali
Finance	_____	employs	_____	Osama Mlkawi
MIS	_____	employs	_____	Ahmad Al ghouli



Cardinality Ratios

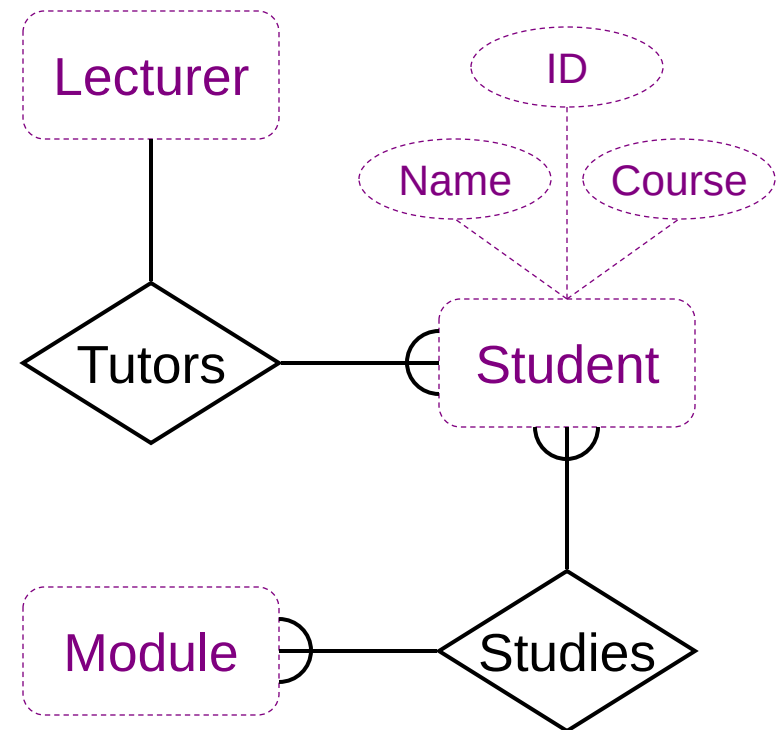
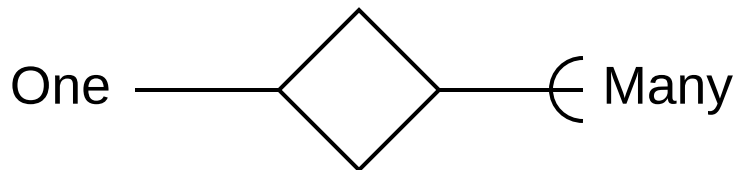
- Each entity in a relationship can participate in zero, one, or more than one instances of that relationship
- This leads to 3 types of relationship...
- One to one (1:1)
 - Each lecturer has a unique office
- One to many (1:M)
 - A lecturer may tutor many students, but each student has just one tutor
- Many to many (M:M)
 - Each student takes several modules, and each module is taken by several students

Cardinality



Diagramming Relationships

- Relationships are links between two entities
- The name is given in a diamond box
- The ends of the link show cardinality

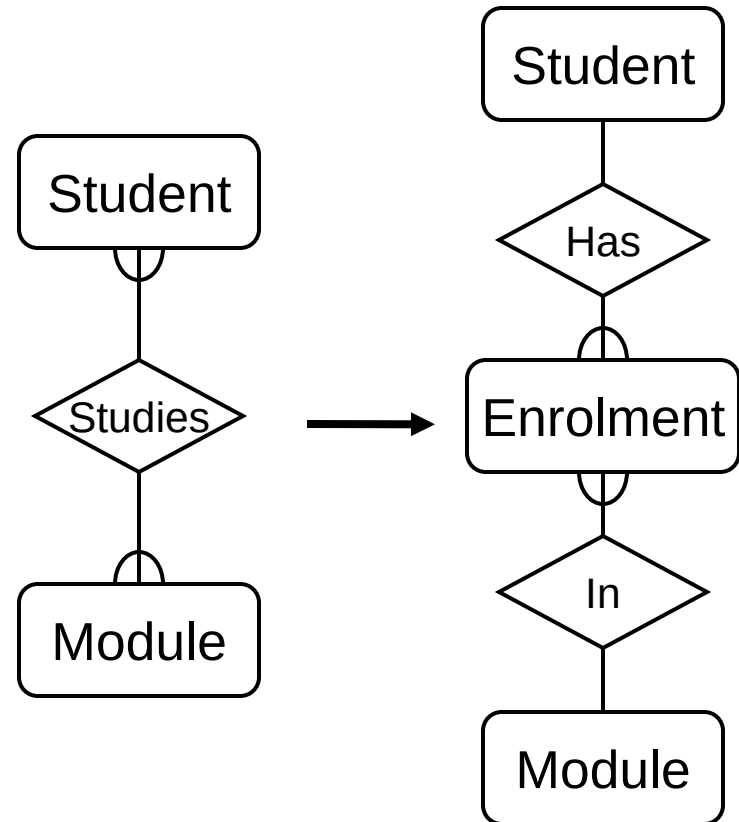


Cardinality: Example

- Customers --> Sales; 1 customer can buy something several times
- Sales --> Customers; 1 sale is always made by 1 customer at the time
- Customers --> Products; 1 customer can buy multiple products
- Products --> Customers; 1 product can be purchased by multiple customers
- Customers --> Shops; 1 customer can purchase in multiple shops
- Shops --> Customers; 1 shop can receive multiple customers
- Shops --> Products; in 1 shop there are multiple products
- Products --> Shops; 1 product (type) can be sold in multiple shops
- Shops --> Sales; in 1 shop multiple sales can be made
- Sales --> Shops; 1 sale can only be made in 1 shop at the time
- Products --> Sales; 1 product (type) can be purchased in multiple sales
- Sales --> Products; 1 sale can exist out of multiple products

Removing M:M Relationships

- Many to many relationships are difficult to represent
- We can split a many to many relationship into two one to many relationships
- An entity represents the M:M relationship



Removing M:M Relationships



Figure 4: Many to many relationship implementation via associative entity.

ERD: Crow's Foot

SYMBOL	MEANING	UML REPRESENTATION
	One and only one	1
	One or many	1..*
	Zero, or one, or many	0..*
	Zero, or one	0..1

Crow's foot notation is a common method of indicating cardinality. The four examples show how you can use various symbols to describe the relationships between entities.

ERD: Chen Notation

Symbol	Notation
	Entity
	Weak Entity
	Relationship Type
	Identifying Relationship Type
	Attribute
	Key Attribute
	Multivalued Attribute
	Derived Attribute
	Composite Attribute
	Total Participation of E2 in R
	Cardinality Ratio 1:N for E1:E2 in R
	Structural Constraint (min,max) on Participation of E in R

Making E/R Models

- To make an E/R model you need to identify
 - Entities
 - Attributes
 - Relationships
 - Cardinality ratios
- from a description
- General guidelines
 - Since entities are things or objects they are often nouns in the description
 - Attributes are facts or properties, and so are often nouns also
 - Verbs often describe relationships between entities

Example

A university consists of a number of departments. Each department offers several courses. A number of modules make up each course. Students enrol in a particular course and take modules towards the completion of that course. Each module is taught by a lecturer from the appropriate department, and each lecturer tutors a group of students

Example - Entities

A university consists of a number of **departments**. Each department offers several **courses**. A number of **modules** make up each course. **Students** enrol in a particular course and take modules towards the completion of that course. Each module is taught by a **lecturer** from the appropriate department, and each lecturer tutors a group of students

Example - Relationships

- A university consists of a number of departments. Each department **offers** several courses. A number of modules **make up** each course. Students **enrol in** a particular course and **take** modules towards the completion of that course. Each module is **taught by** a lecturer **from the** appropriate department, and each lecturer **tutors** a group of students

Example - E/R Diagram

Entities: Department, Course, Module, Lecturer, Student

Department

Course

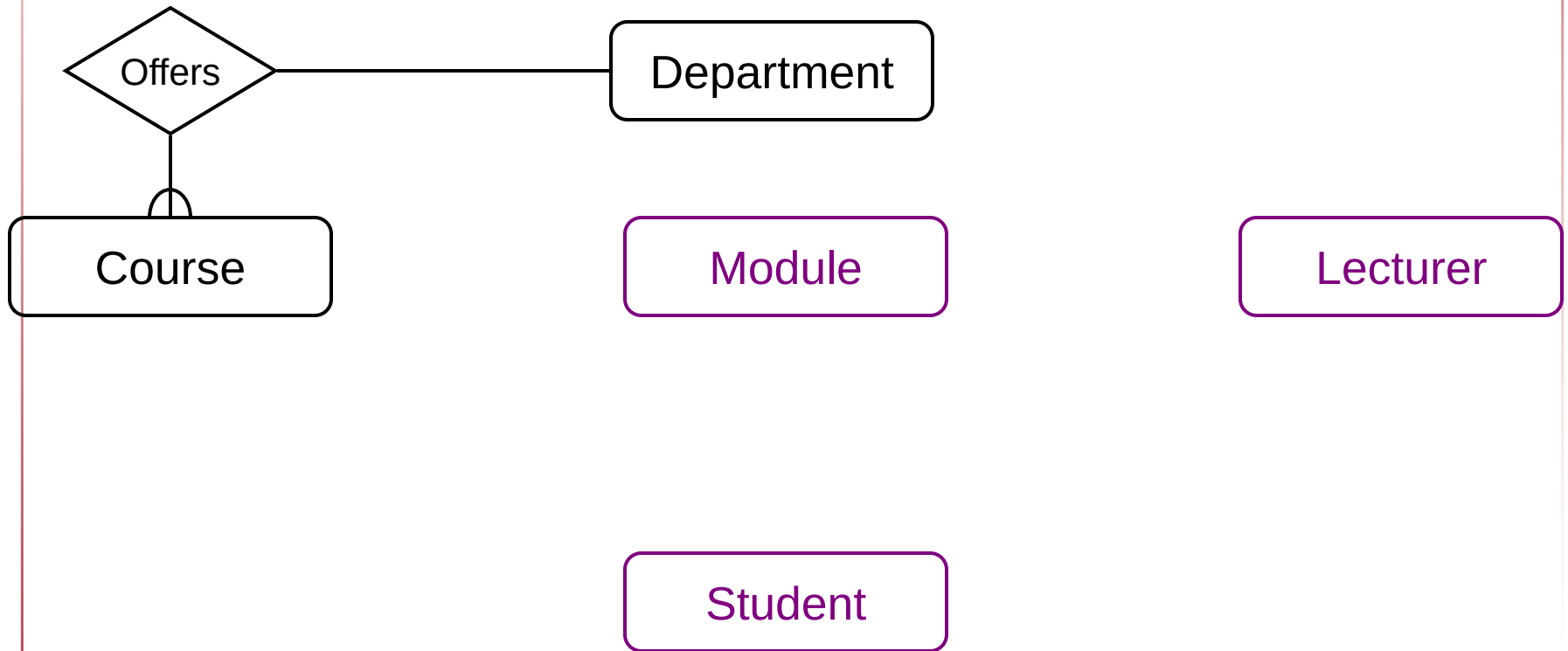
Module

Lecturer

Student

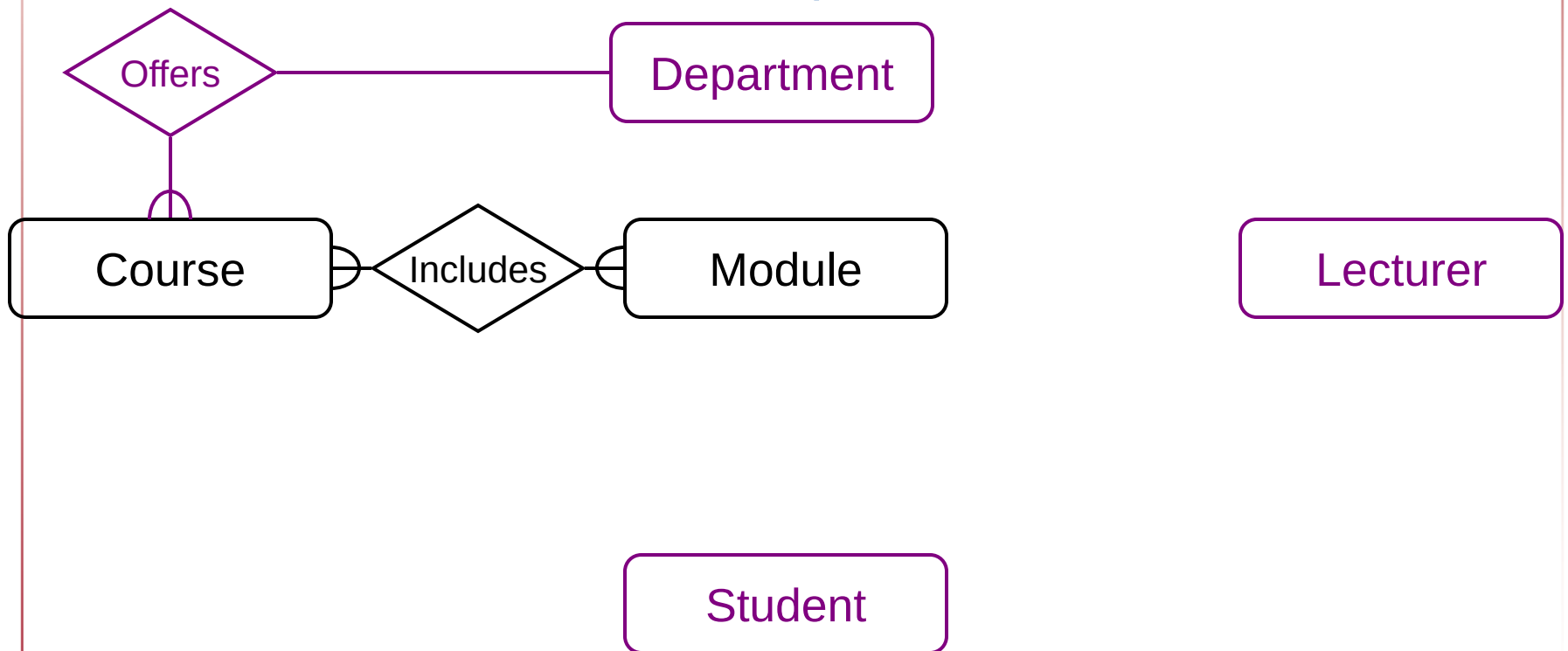
Example - E/R Diagram

Each department **offers** several courses



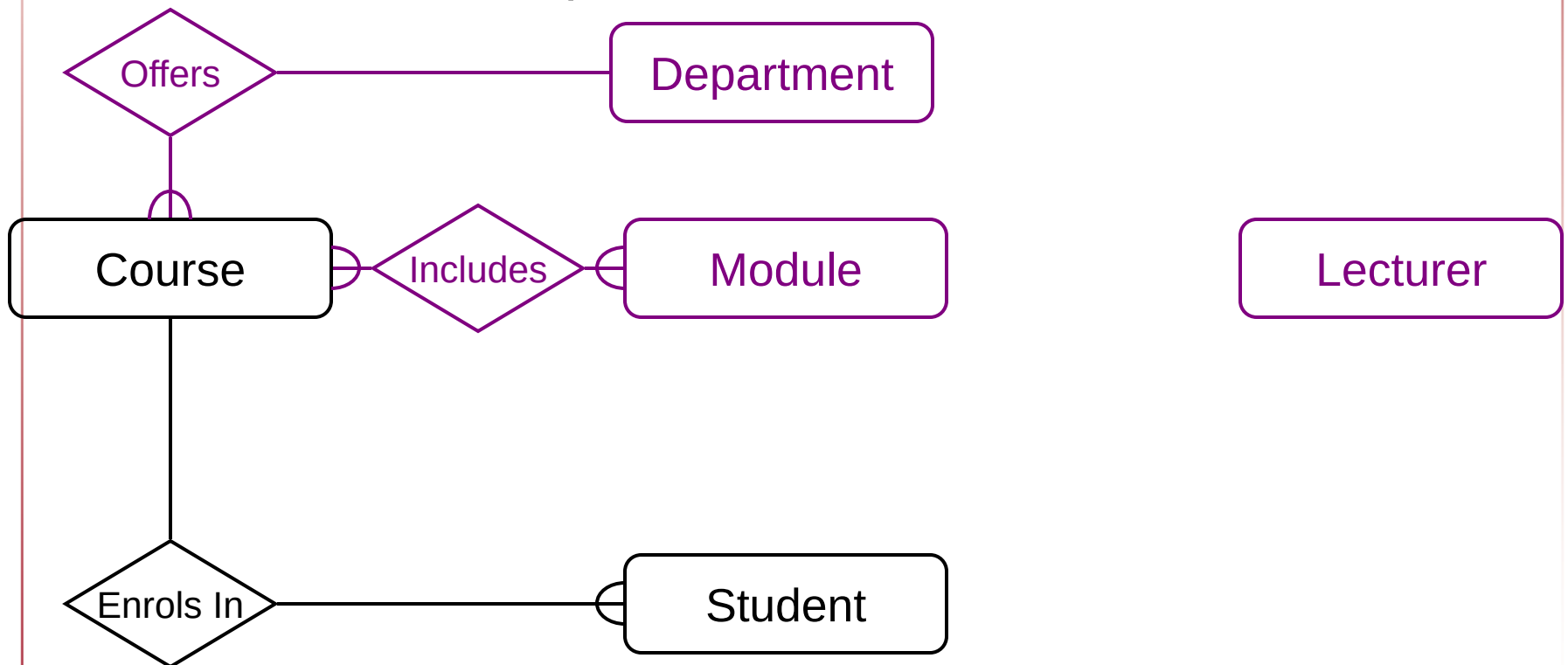
Example - E/R Diagram

A number of modules **make up** each courses



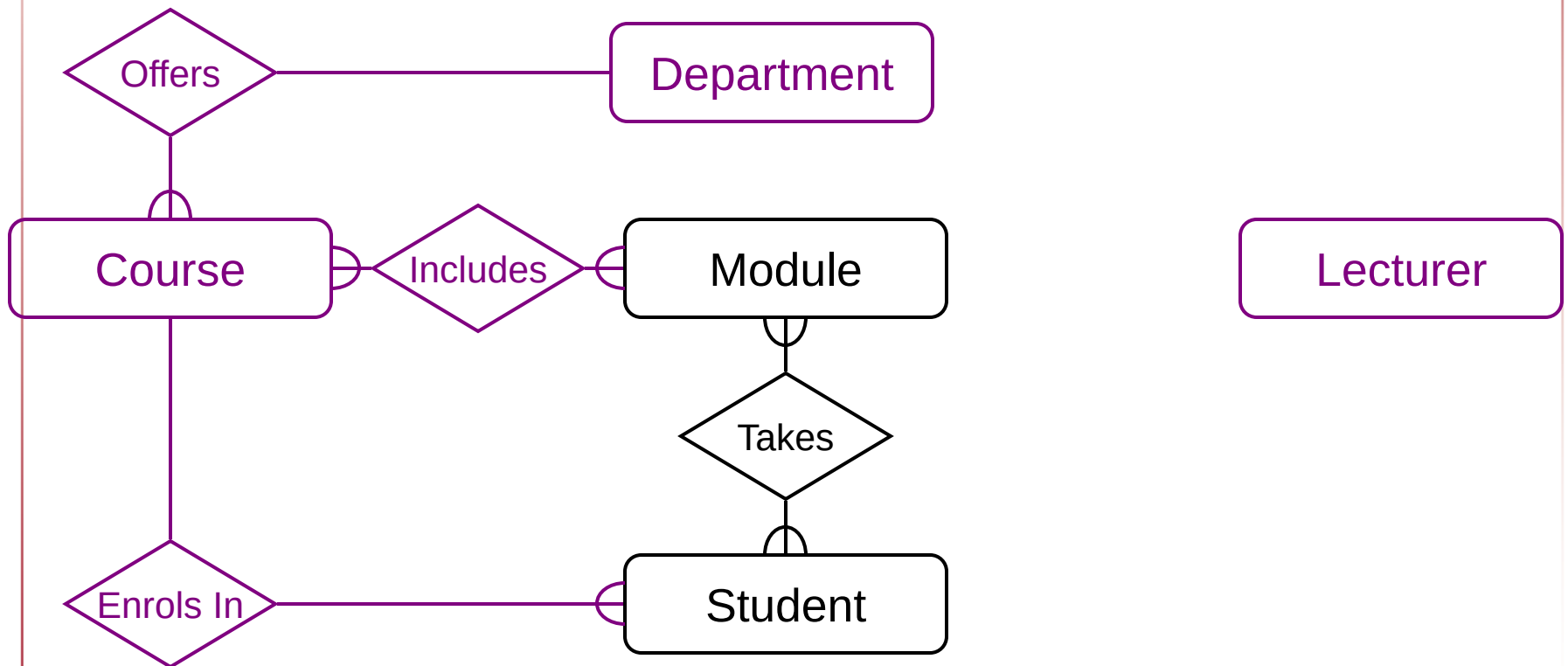
Example - E/R Diagram

Students **enrol in** a particular course



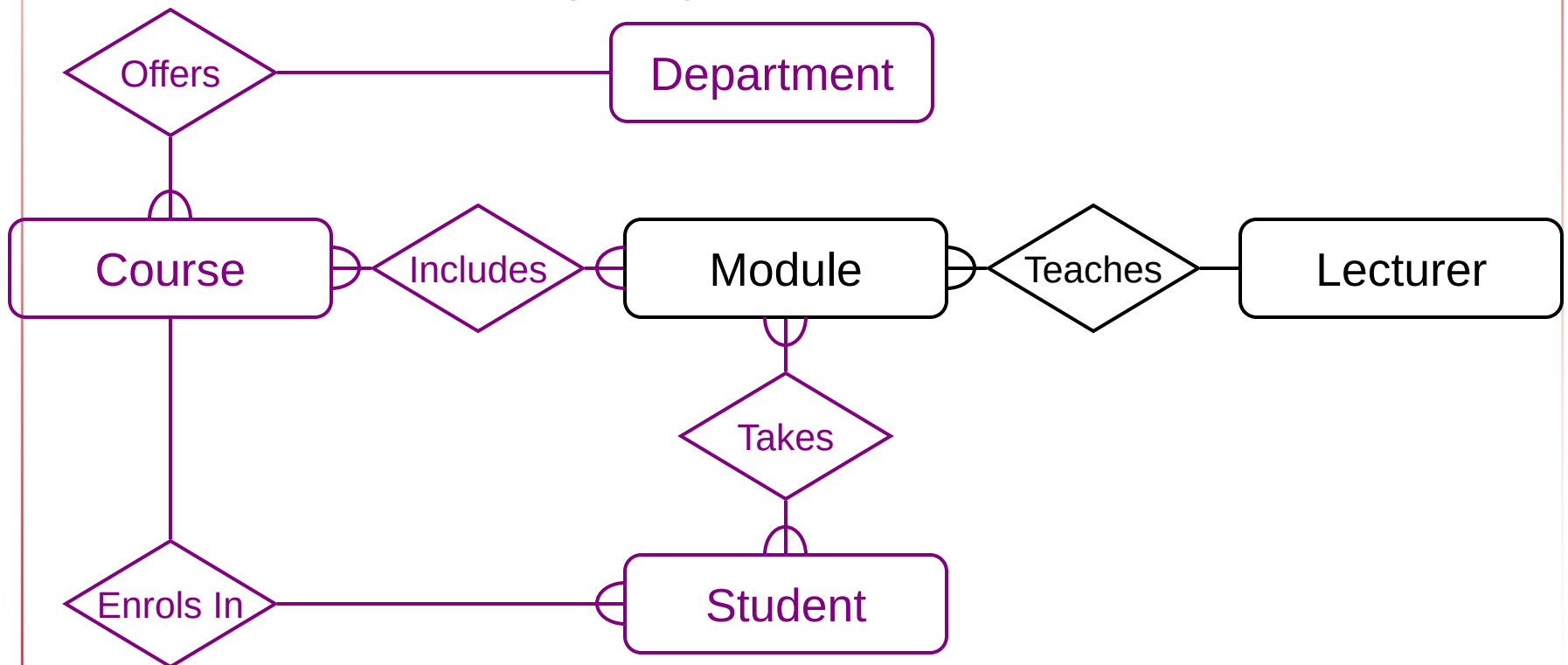
Example - E/R Diagram

Students ... **take** modules



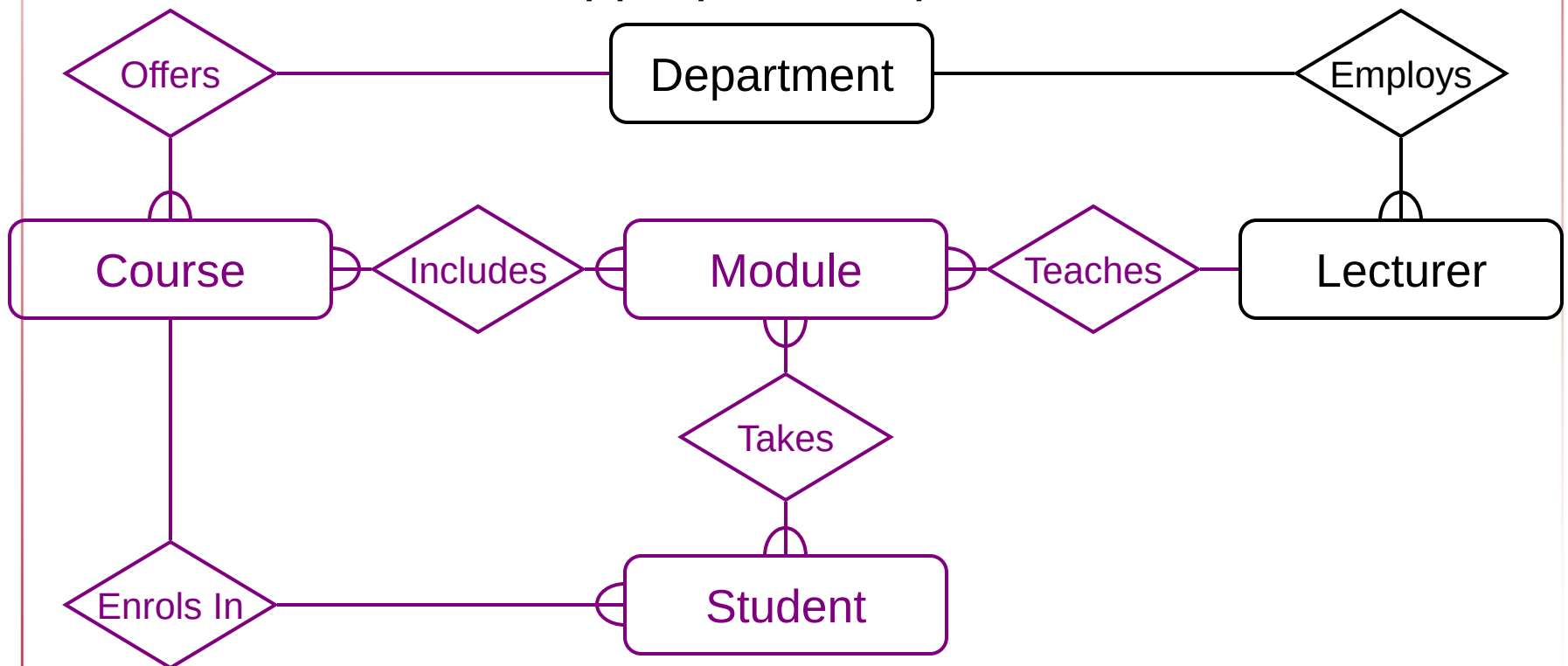
Example - E/R Diagram

Each module is taught by a lecturer



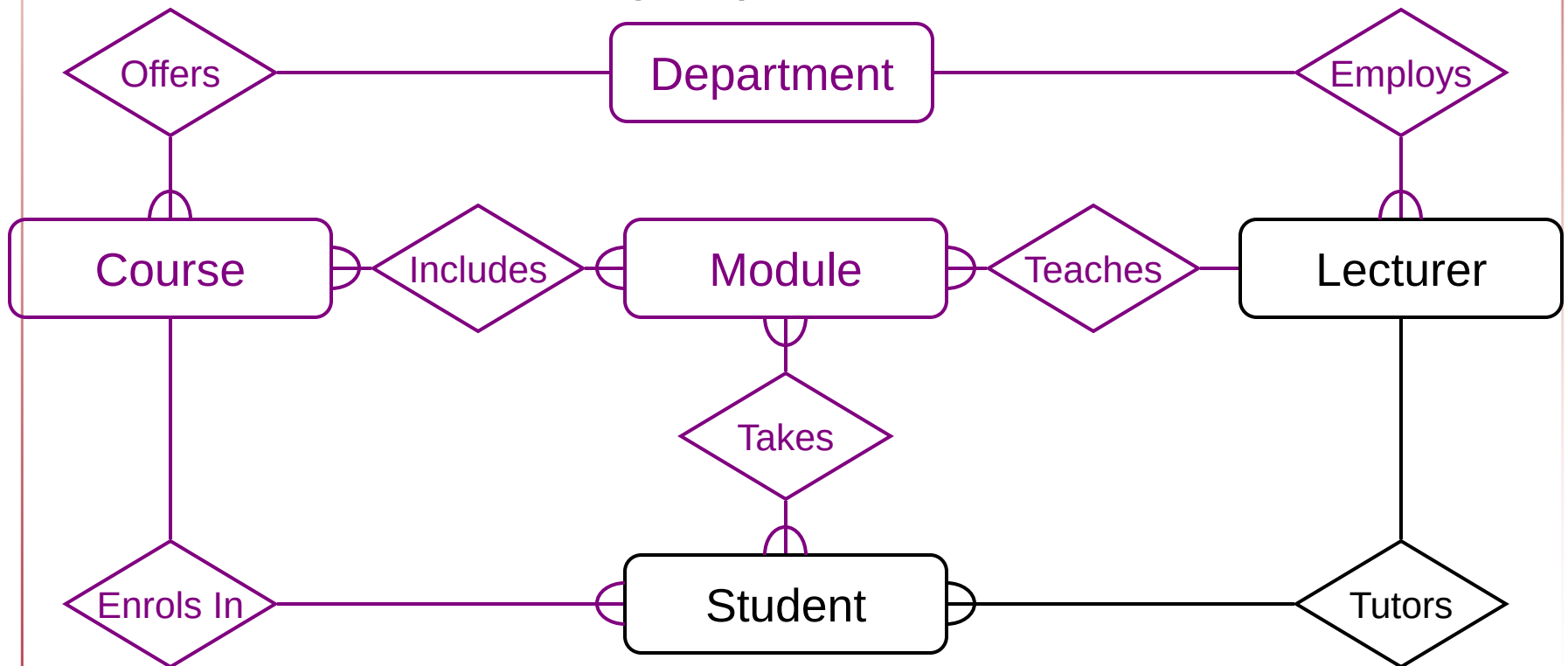
Example - E/R Diagram

a lecturer **from the** appropriate department

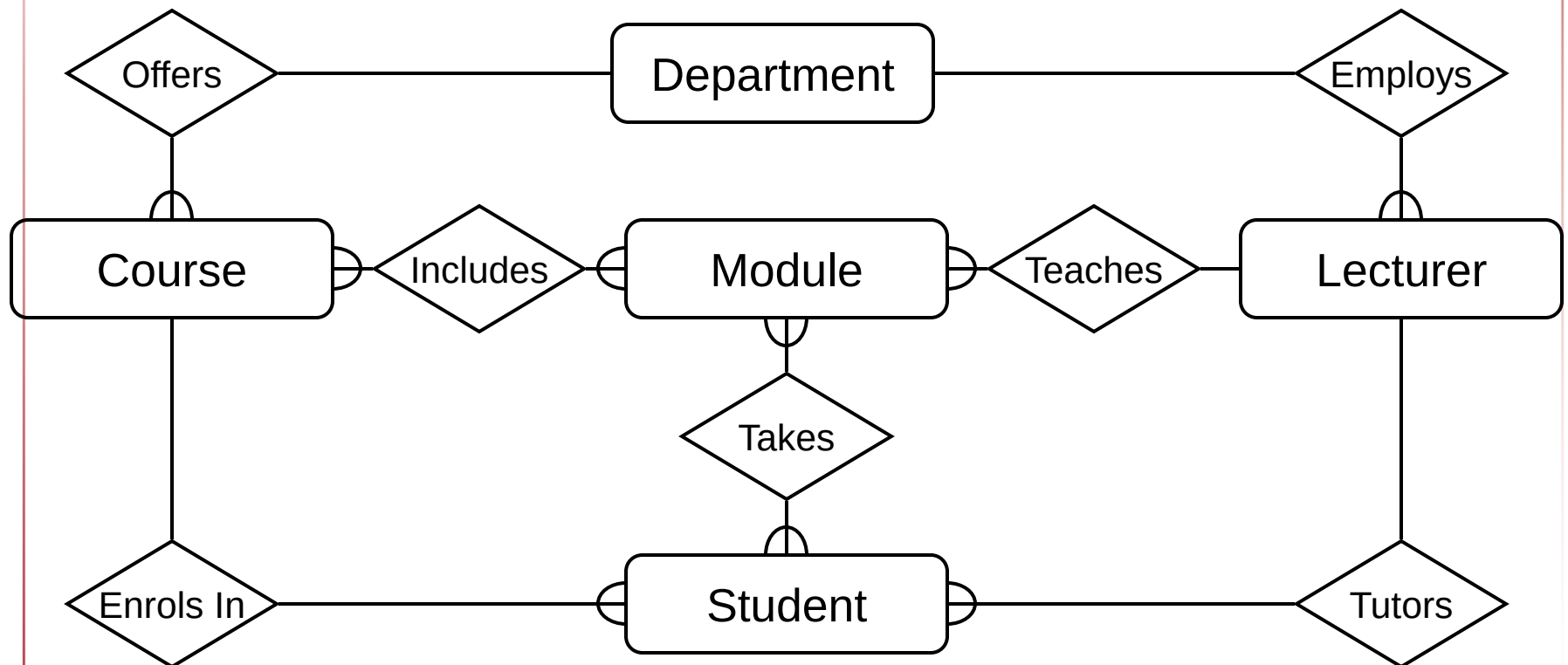


Example - E/R Diagram

each lecturer **tutors** a group of students



Example - E/R Diagram



Entity-Relationship Diagrams

In the first example of cardinality notation, one and only one CUSTOMER can place anywhere from zero to many of the ORDER entity.

In the second example, one and only one ORDER can include one ITEM ORDERED or many.

In the third example, one and only one EMPLOYEE can have one SPOUSE or none.

In the fourth example, one EMPLOYEE, or many employees, or none, can be assigned to one PROJECT, or many projects, or none.

EXAMPLES OF CARDINALITY NOTATION



One and only one CUSTOMER can place anywhere from zero to many of the ORDER entity.



One and only one ORDER can include one ITEM ORDERED or many.



One and only one EMPLOYEE can have one SPOUSE or NONE.



One EMPLOYEE, or many employees, or none, can be assigned to one PROJECT, or many projects, or none.

Example ERD

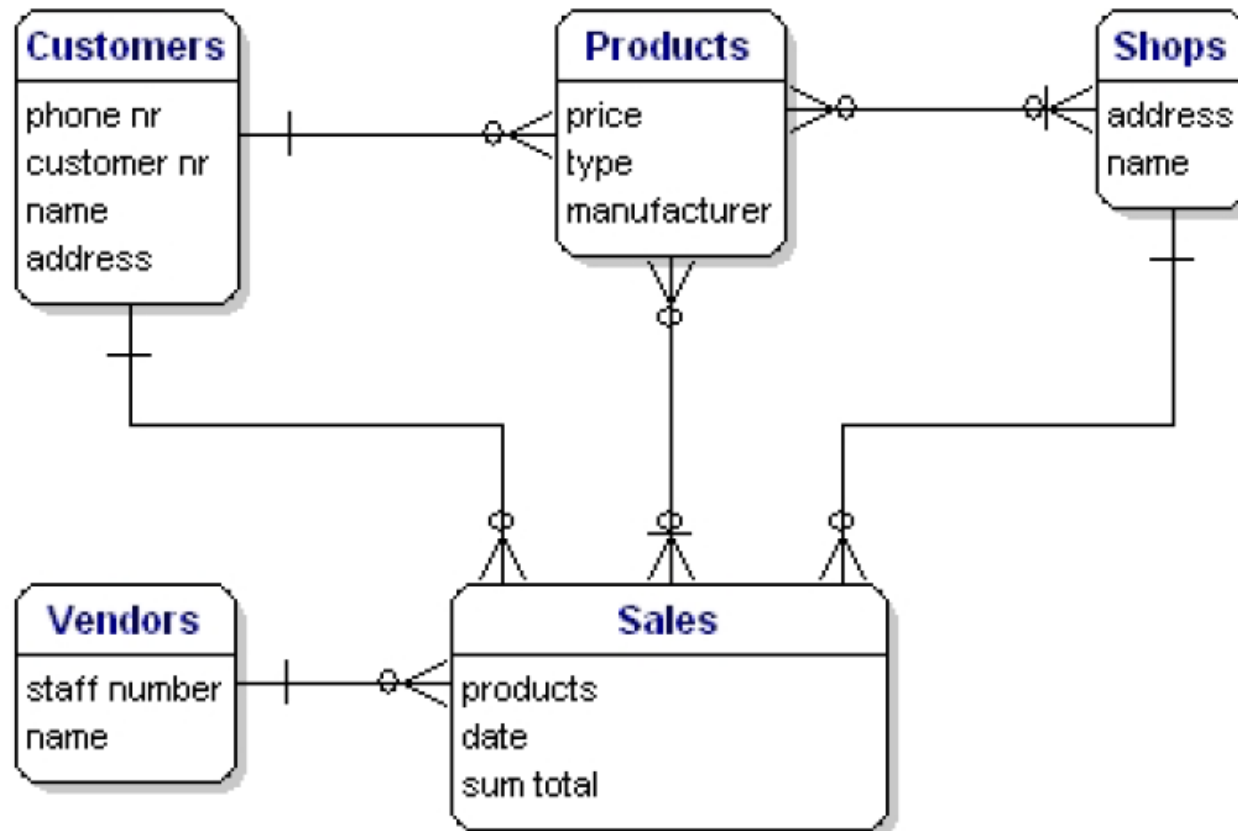


Figure 2: Relationships between the entities.

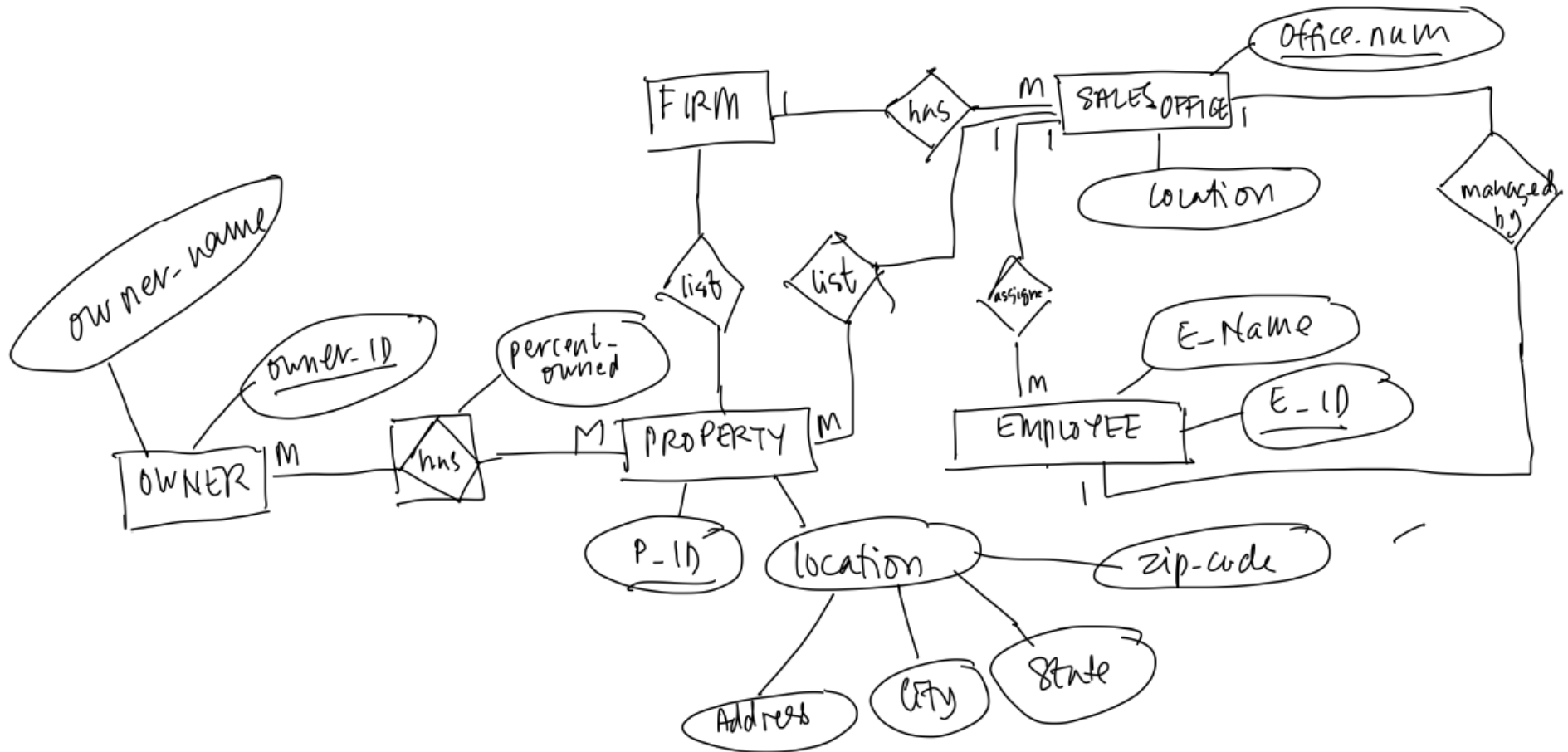
ERD EXERCISE 1

A university consists of a number of departments. Each department offers several courses. A number of modules make up each course. Students enrol in a particular course and take modules towards the completion of that course. Each module is taught by a lecturer from the appropriate department, and each lecturer tutors a group of students

(erd2)

ERD EXERCISE 2

- Prepare an ERD for a real estate firm that lists property for sale. The following describes this organization:
- The firm has a number of sales offices in several states. Attributes of sales office include Office_Number(identifier) and Location
- Each sales office is assigned one or more employees. Attributes of employee include Employee_ID(identifier) and Employee_Name. An employee must be assigned to only one sales office.
- For each sales office, there is always one employee assigned to manage that office. An employee may manage only the sales office to which he/she is assigned.
- The firm lists property for sale. Attributes of property include Property_ID(identifier) and Location. Components of Location include Address, City, State and Zip_Code.
- Each unit of property must be listed with one (and only one) of the sales offices. A sales office may have any number of properties listed, or may have no properties listed.
- Each unit of property has one or more owners. Attributes of owners are Owner_ID(identifier) and Owner_Name. An owner may own one or more units of property. An attribute of the relationship between property and owner is Percent_Owned.



Exercise 3

Pick and Shovel Construction Company is a multi-state building contractor specializing in medium-priced town homes. Assume that Pick and Shovel's main entities are its customers, employees, projects and equipment. A customer can hire the company for more than one project, and employees sometimes work on more than one project at a time. Equipment, however, is assigned to only one project at a time. Draw a complete ERD showing those entities.

Exercise 3 (in slides)

